Atomic Physics

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Atomic Structure

INTRODUCTION:

Atom is a Greek word, and its meaning is indivisible, which is equal to an ultimate particle that cannot be further subdivided. John Dalton considered that all matter is composed of the smallest indivisible particle called an atom.

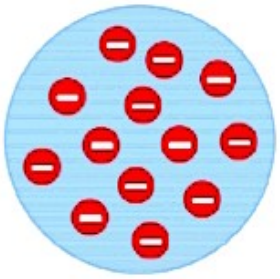
'In this chapter, we will discuss different atomic models of past two centuries and will discuss one of the important atomic models that is Bohr's atomic model of the atom'. All these atomic models are based on Newton's classical mechanics. The atom as we understand it today is based on Quantum Mechanics.

Atomic Models:

Thomson's Model of Atom (1904)

Proposed by Joseph J. Thomson in 1904.

The model suggested that the atom is a region in which a positive charge is spread out in space with electrons embedded throughout the region just like seeds in a watermelon.



This model knows as the watermelon or plum pudding model.

Drawbacks:

1. The mass of an atom is uniform this was the wrong concept
2. It is not reflected the movement of electrons.
3. This model couldn't explain the stability of atom.

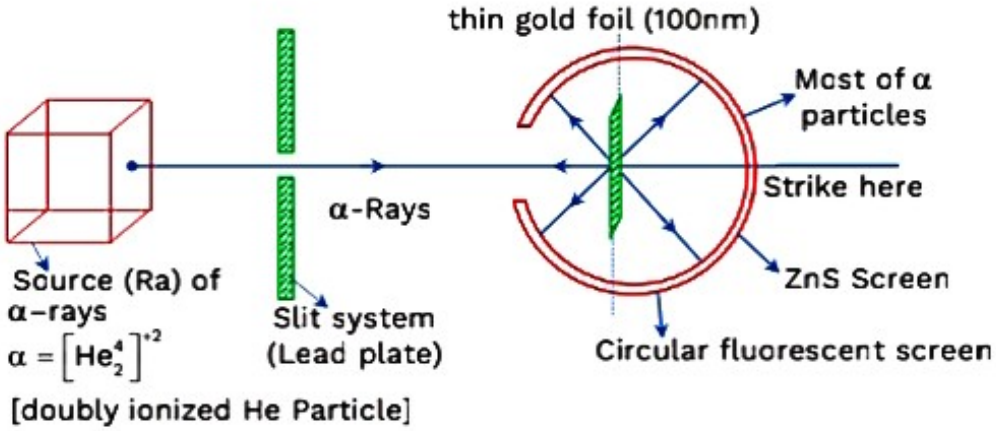
Rutherford's Atomic Model (α-particle experiment) :-

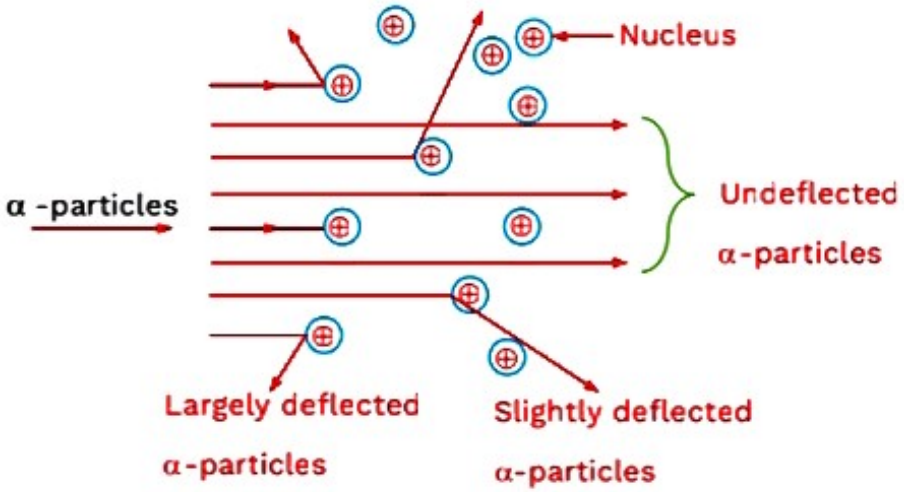
In 1911, Earnest Rutherford performed an experiment in which a beam of positively charged. α-particle was projected into a 100 nm thin gold foil.

Experimental observation and conclusion:-

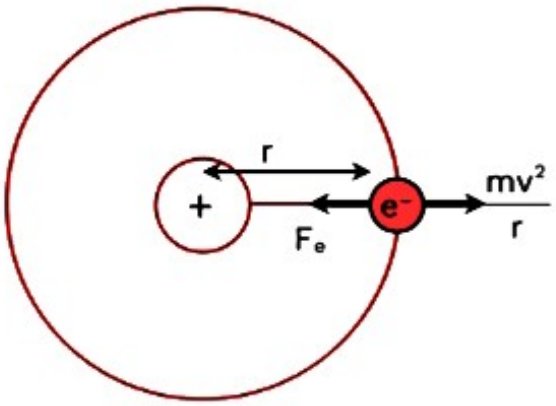
(1) Maximum number of α-particle goes undeviated. On the basis of this, he concluded that most spaces of an atom are empty.

(2) Approx out of 20,000 alpha particles only one or two α-particle deviated by 180°. He concluded that most of the space in a region is empty and there is a very small positive charged sphere at the center called nucleus and electron is revolving around it.





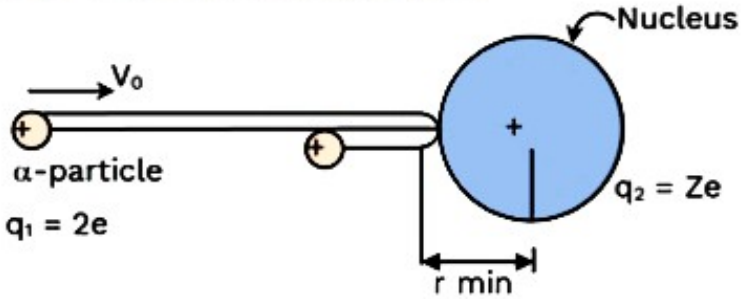
(3) The centripetal force of electron is provided by the Lorentz force between positive charge at the center and electron.



$$F_e = \frac{m_e v^2}{r}$$

$$\frac{K(Ze)(e)}{r^2} = \frac{mv^2}{r}$$

(4) For finding the radius of the nucleus he bombarded high energy α-particle and calculated the minimum distance of the closest approach.



Apply energy conservation

Kinetic energy = Electrostatic potential energy

$$\frac{1}{2}m_\alpha V_0^2 = \frac{K(Ze)(2e)}{r_{min^2}}$$

From above relation we can calculate radius of nucleus. Radius of nucleus is in order of $\approx 1 \times 10^{-15}$ meter.

According to Rutherford Scattering Formula:

The number of α-particle scattered at an angle θ by a target are given by:

$$N_0 = \frac{N_0 nt\left(2Ze^2\right)^2}{4\left(4\pi \epsilon_0\right)^2 r^2 \left(mv_0^{\,2}\right)^2} \times \frac{1}{\sin^4 \theta/2}$$

Where N_0 = number of α-particle that strike the unit area of the scatter

n = number of target atom per m^3

t = thickness of target nucleus

Ze = charge on target nucleus

2e = charge on α particles

r = distance of the screen from target

v_0 = Initial velocity of α-particles

$\Rightarrow$ **During the experiment, the following conclusions were made.**

(i) If ϕ is the angle made by a scattered particle with its original direction of motion and N is the number of particles available in that direction of motion, it was found that

$$\frac{1}{\sin^4(\phi/2)} \propto N$$

(ii) If t is the thickness of the foil and N is the number of α-particles scattered in a particular direction (ϕ = constant), it was observed that

$$\frac{N}{t} = \text{constant}$$

$$\frac{N_1}{N_2} = \frac{t_1}{t_2}$$

Q. **In Rutherford's scattering experiment, if the number of α-particles scattered at an angle of 90° is 200, then find out the number of α-particles scattered at an angle of 60°.**

Sol. We know that:

$$N \propto \frac{1}{\sin^4\left(\frac{\phi}{2}\right)}$$

$$\frac{N_{60°}}{N_{90°}} = \frac{\sin^4(90°/2)}{\sin^4(60°/2)} = \frac{\sin^4(45°)}{\sin^4(30°)}$$

$$= \left(\frac{1}{\sqrt{2}}\right)^4 / \left(\frac{1}{2}\right)^4 = 4$$

$$\frac{N_{60°}}{N_{90°}} = \frac{N_{60°}}{55} = 4$$

$$N_{60°} = 4 \times 200 = 800$$

Q. **A Head-on collision takes place between an α-particle of kinetic energy 5.5 MeV and a gold nucleus (Z = 79), calculate the distance of the closest approach.**

Sol. Since the distance of the closest approach is given by

$$r_0 = \frac{1}{4\pi\epsilon_0}\left(\frac{2Ze^2}{m_\alpha V_\alpha^2}\right) = \frac{1}{4\pi\epsilon_0}\left(\frac{2Ze^2}{K_\alpha}\right)$$

$$r_0 = 9 \times 10^9 \times \frac{2 \times 79 \times (1.6 \times 10^{-19})^2}{(5.5 \times 1.6 \times 10^{-13})}$$

$$r_0 = 4.13 \times 10^{-14}\ m$$

The radius of the gold nucleus must be smaller than r_0, so it may be lie between 10^{-14} to 10^{-15} m.

The trajectory of an Alpha Particle and Impact Parameter:

When an alpha particle passes through 1222the gold foil, the nucleus starts repulsing the alpha particle due to this alpha particle getting deflected from an initial direction, the angle between final velocity and initial velocity vector of is θ, and the perpendicular distance of initial velocity vector of an alpha particle from nucleus called impact parameter(b).



The angle between the direction of approach of the α-particle and direction in which it finally goes is defined as the angle of scattering and is defined by θ.

Rutherford concluded that:

$$\tan\left(\frac{\theta}{2}\right) = \frac{1}{b}\left(\frac{Ze^2}{4\pi\epsilon_0 K_\alpha}\right)$$

$$b = \frac{Ze^2}{4\pi\epsilon_0 K_\alpha}\cot\left(\frac{\theta}{2}\right)$$

$$K_\alpha = \text{constant}$$

$$b \propto \cot\left(\frac{\theta}{2}\right)$$

A graph between b and $\cot\left(\frac{\theta}{2}\right)$ is a straight line.

Q. **Let a 5MeV α particle is scattered by 74° when it approaches a gold nucleus (Z = 79). Find the impact parameter.**

Sol. Since, Z = 79

K_α = 5MeV = $5 \times 1.6 \times 10^{-13}$ J

$= 8 \times 10^{-13}$ J

$\theta = 74°$

$$\Rightarrow \quad \frac{\theta}{2} = 37°$$

$$\cot\left(\frac{\theta}{2}\right) = \frac{4}{3}$$

Using the Relation

$$\tan\left(\frac{\theta}{2}\right) = \frac{1}{b}\left(\frac{Ze^2}{4\pi \epsilon_0 K_\alpha}\right)$$

$$b = \frac{Ze^2}{4\pi \epsilon_0 K_\alpha} \cot\left(\frac{\theta}{2}\right)$$

$$b = \frac{79 \times \left(1.6 \times 10^{-19}\right)^2 \times 9 \times 10^9 \times \frac{4}{3}}{8 \times 10^{-13}}$$

$$\boxed{b = 3.03 \times 10^{-14} m}$$

The Bohr's Atomic Model for Hydrogen atom:

In 1911, Niels Bohr's use quantum theory and proposed a quantum mechanical atomic model for Hydrogen atom. It is based on the following three postulates:-

(1) **First-Postulate:-** Bohr's first postulate was that an electron in an atom revolves in certain stable orbits without the emission of radiant energy contrary to the prediction of electromagnetic theory.

According to this postulates, **'Each atom has certain definite stable state in which it can exist and each possible state has definite total energy'**. There are called the stationary state of the atom.

$$\frac{mv^2}{r} = \frac{e^2}{4\pi \epsilon_0 r^2} \qquad ...(1)$$

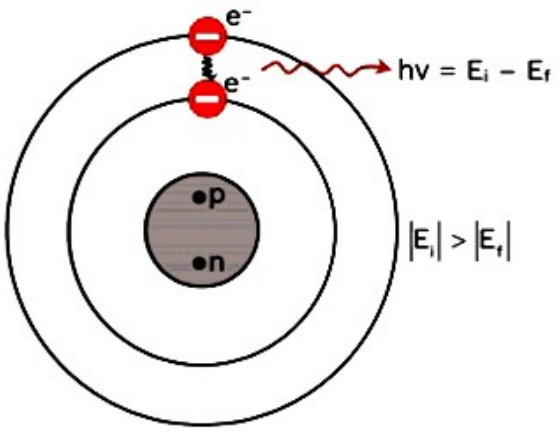
(2) **Second Postulate:-** This postulate states that the electron revolves around the nucleus only in those orbits for which the angular momentum is some integral multiple of $(h/2\pi)$ where h is the plank's constant.

$$\vec{L} = n \times \left(\frac{h}{2\pi}\right) \qquad ...(2)$$

where, n = principle quantum number.

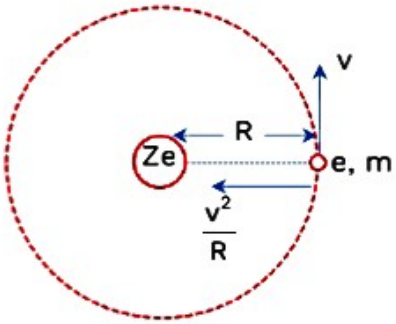
(3) **Third Postulate:-** Bohr's third postulate incorporated into atomic theory and early quantum concept that had been developed by Plank and Einstein.

It states that an electron might make a transition from one of its specific non-radiating orbits to another of lower energy. When it does so, a photon is emitted having energy equal to the energy difference between the initial and final states.



Mathematical Relation

(1) Calculation of Radius :-



Electrostatic force of attraction is balanced by centripetal force:

$$\frac{K(Ze)(e)}{r^2} = \frac{mv^2}{r}$$

$$\frac{KZe^2}{r^2} = \frac{mv^2}{r} \qquad \ldots(1)$$

By using Second postulates :-

$$mvr = n \times \frac{h}{2\pi} \qquad \ldots(2)$$

From equation (1) and (2)

$$\frac{KZe^2}{r^2} = \frac{m}{r} \times \frac{n^2h^2}{4\pi^2m^2r^2}$$

$$r = \frac{m_e \times n^2 \times h^2}{KZe^2 4\pi^2 m_e^2}$$

$$r = \frac{h^2}{4\pi^2 m_e K Z_e^2}\left(\frac{n^2}{Z}\right)$$

Put the value of h, m_e, K, e and calculate the value of r.

$$\boxed{r = 0.529 \times \frac{n^2}{Z}\text{ Å}}$$

The smallest radius corresponds to n = 1.

$$r = 0.529 \times \frac{1^2}{1}$$

$$\boxed{r_0 = 0.529\text{ Å}}$$

↓

Called Bohr Radius.

Note : Bohr's model is only valid for H-like atom

For example : He^{+} ; Z = 2

Li^{++} ; Z = 3

Be^{+++} ; Z = 4

For the above atoms use the formula for the calculation of the radius of the nth orbit.

$$\boxed{r_n = 0.529 \times \frac{n^2}{Z}\text{ Å}}$$

(2) Velocity calculation :-

From equation (2)

$$v_n = \frac{n \times h}{2\pi m_e \times \left\{0.529 \times \frac{n^2}{Z}\text{ Å}\right\}}$$

$$v_n = 2.18 \times 10^6 \left(\frac{Z}{n}\right)\text{m/sec}$$

So, velocity of electron in n^{th} orbit :-

$$\left\{\begin{matrix} v_n \propto Z \\ v_n \propto \frac{1}{n} \end{matrix}\right\}$$

Angular Velocity of Electron in n^{th} orbit:

Use relation between angular velocity and linear velocity.

$$\omega_n = \frac{v_n}{r_n}$$

$$\omega_n = \frac{8\pi^3 K^2 Z^2 e^4 m}{n^3 h^3}$$

$$\omega_n \propto \frac{Z^2}{n^3}$$

Frequency of Electron in n^{th} orbit:

$$f_n = \frac{\omega_n}{2\pi}$$

$$f_n = \frac{4\pi^2 K^2 Z^2 e^4 m}{n^3 h^3}$$

$$\boxed{f_n \propto \frac{Z^2}{n^3}}$$

The time Period of Electron revolving in n^{th} (Bohr's) orbit:

$$T_n = \frac{1}{f_n}$$

$$\boxed{T_n = \frac{n^3 h^3}{4\pi^2 K^2 Z^2 e^4 m}}$$

$$\boxed{T_n \propto \frac{n^3}{Z^2}}$$

Current in n^{th} 'Bohr's orbit:

$$I_n = f_n \times e$$

$$I_n = \frac{4\pi^2 K^2 Z^2 e^5 m}{n^3 h^3}$$

$$\boxed{I_n \propto \frac{Z^2}{n^3}}$$

Magnetic Induction at the nucleus to n^{th} orbit:

$$\text{Magnetic Induction} = \frac{\mu_0 I_n}{2r_n}$$

$$B_n = \frac{8\pi^4 \mu_0 K^3 Z^3 e^7 m^2}{n^5 h^5}$$

$$\boxed{B_n \propto \frac{Z^3}{n^5}}$$

Magnetic Moment of the nth Bohr's Orbit:

The magnetic moment of the nth Bohr's orbit due to the electron is given by:

$$M_n = I_n \times A_n$$

$$M_n = I_n \times \pi r_n^2$$

$$= \frac{4\pi^2 K^2 Z^2 e^5 m}{n^3 h^3} \times \pi \times \left(\frac{n^2 h^2}{4\pi^2 K Z e^2 m}\right)^2$$

$$M_n = \frac{enh}{4\pi m}$$

In this equation for n = 1, the magnetic moment is also known as Bohr's magnetron.

The energy of an electron in n^{th} orbit:

Now we can calculate total energy of electron revolving in n^{th} orbit .Total energy is sum of kinetic and potential energy:

$$(\text{Total Energy})_n = (\text{Kinetic energy})_n + (\text{Potential Energy})_n \quad \ldots(1)$$

The Kinetic Energy of an electron in the nth orbit can be given as

$$K.E_n = \frac{1}{2}mV_n^2 \quad \ldots(2)$$

From equation of first postulate of Bohr model, for n^{th} orbit

$$mV_n^2 = \frac{KZe^2}{r_n} \quad \ldots(3)$$

From (2) & (3) we have

$$KE_n = \frac{1}{2}\frac{KZe^2}{r_n} \quad \ldots(4)$$

The potential energy of an electron in the nth orbit is given as

$$U_n = -\frac{KZe^2}{r_n}$$

Thus, total energy of e^- in n^{th} orbit can be given as

$$E_n = K.E_n + P.E_n$$
$$= KE_n + U_n$$

$$E_n = \frac{1}{2}\frac{KZe^2}{r_n} - \frac{KZe^2}{r_n}$$

$$\boxed{E_n = -\frac{1}{2}\frac{KZe^2}{r_n}}$$

Important Point: From above results we can observe that $|E_n| = |KE_n| = \frac{1}{2}|U_n|$. This relation is valid for all those particle which revolinge under the action of inverse square law.

Total Energy of e^- in n^{th} Orbit:

$$E_n = -\frac{1}{2}\frac{KZe^2}{r_n} \quad \ldots(1)$$

Put the value of r_n in Eq. (1)

$$E_n = -\frac{1}{2}KZe^2 \times \frac{4\pi KZe^2 m}{n^2h^2}$$

$$E_n = -\frac{2\pi^2K^2Z^2e^4m}{n^2h^2}$$

$$E_n = -\frac{2\pi^2K^2e^4m}{h^2}\frac{Z^2}{n^2}$$

Put the value of the K,e,m,h, we get

$$E_n = -13.6 \times \frac{Z^2}{n^2} \text{ eV}$$

The above equation can be used to find out the energies of electrons in the different energy levels of different Hydrogen like atoms.

Energies of Different Level in Hydrogenic Atoms:

Now use the above equation $E_n = -13.6 \times \frac{Z^2}{n^2}$ eV , we can calculate the energy level of hydrogen or hydrogen like atom.

$E_1 = -13.6\ Z^2$ eV
$E_2 = -3.4\ Z^2$ eV
$E_3 = -1.51\ Z^2$ eV
⋮ ⋮ ⋮

We can clearly observe that as n increases the energy gap between consecutive states decreases.

The Energy Level Diagram for a Hydrogen Atom:



Q. **Hydrogen gas is in the atomic state and it is in excited energy level so that its electrostatic potential energy is equal to -1.7eV. Suppose a photoelectric plate having work function 2.3 eV is exposed to the emission spectra of the gas. Assuming all the transitions to be possible, Calculate the least de-Broglie wave-length of emitted photoelectrons.**

Sol. Given that electrostatic potential energy of H atom V

PE = −1.7 eV

Since we know that kinetic energy is given by

$$K.E = \left|\frac{P.E}{2}\right| = \frac{1.7}{2} = 0.85\,eV$$

So, total energy is E = −1.7 + 0.85 = −0.85 eV

Since, $E_n = \frac{-13.6}{n^2} = -0.85\,eV$

$$E_n = \frac{-13.6}{n^2} = -0.85\,eV$$

$$n^2 = \frac{-13.6}{0.85} = 16 \Rightarrow n = 4$$

So, the atom excited to n = 4 states, the maximum energy will be emitted when electrons will make the transition from n = 4 to n = 1 the energy emitted for this transition.

$$\Delta E = -0.85 - (-13.6) = 12.75eV$$

When a photon of this energy is incident on a metal plate having work function 2.3 eV, the kinetic energy of the fastest electron ejected will be given by

$$K_{max} = \Delta E - \phi_0 = 12.75 - 2.3 = 10.45\,eV$$

The minimum de-Broglie wave-length is given by:

$$\lambda_{min} = \frac{h}{P_{max}} = \frac{h}{\sqrt{2mK_{max}}}$$

$$\lambda_{min} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times (9.1 \times 10^{-31})(10.45 \times 1.6 \times 10^{-19})}}$$

$$\lambda_{min} = 3.8 \times 10^{-10} m$$
$$= 3.8\ \text{Å}$$

Some Standard Definitions:-

(1) Ground State:- The energy of an atom is the least when its electron is revolving in an orbit closest to the nucleus i.e. for which n = 1.

The lowest state of the atom is called the ground state, this state has the lowest energy. The energy of this state is given below for different atoms

$$\text{Energy } E_n = -13.6 \times \frac{Z^2}{n^2} eV$$

(i) For H atom ; Z = 1 ; n = 1 ; E = –13.6 eV.

(ii) For He^+ Ion ; Z = 2 ; n = 1 ; E = – 54.4 eV.

(iii) For Li^{+++} Ion ; Z = 3 ; n = 1 ; E = –112.4 eV.

(2) Excited State:- All other states except the ground state.

n = 2, for first excited state

n = 3, for second excited state

where n = principle quantum number.

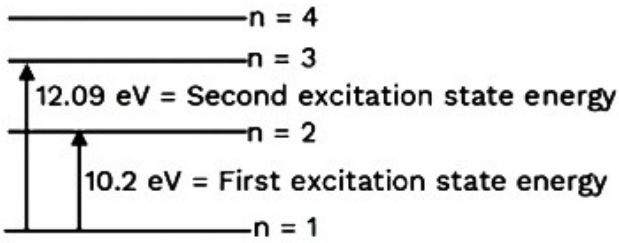
(3) Ionisation Energy:- Energy required to take an electron from ground state to infinity called ionization energy.

$$E_n = -13.6 \times \frac{Z^2}{n^2} eV$$

For H-atom ; E = –13.6 eV ; I.E. = +13.6 eV

For He-atom ; E = –54.4 eV ; I.E. = +54.6 eV

(4) Excitation-Energy:- The energy required of an electron to move from the ground state of an atom to another state of the atom, called to excitation energy of that state.



Excitation Energy of H-atom

(5) Ionisation Potential (I.P.):- Potential difference by which an electron should be accelerated so that it can make the transition from ground state to infinity.
Ionisation Potential of Hydrogen atom = 13.6 eV
Ionisation Potential of He-Ion = 54.4 volt.

Energies of Different Level in Hydrogenic Atoms:

By the use of equation $E_n = -13.6 \times \frac{Z^2}{n^2}$ eV, we can find out the energies of different energy levels students should remember these energies for first six levels as

$E_1 = -13.6\ Z^2$ eV

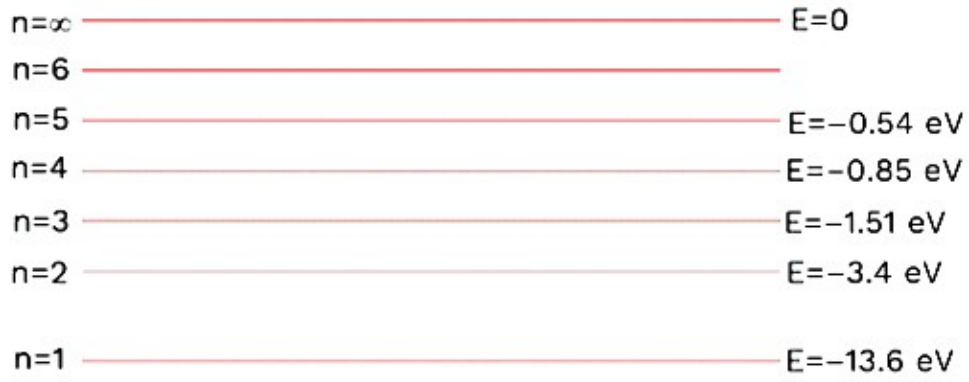
$E_2 = -3.4\ Z^2$ eV

$E_3 = -1.51\ Z^2$ eV

⋮ ⋮ ⋮

Equation clearly shows that as the value of 'n' increases, the difference between two consecutive energy levels decreases.

The Energy Level Diagram for a Hydrogen Atom:



Q. **The electron in a Hydrogen atom makes transition state n_1 to state n_2. Assume Bohr's atomic model is valid. It is given that, the time period of an electron in the initial state is eight times that final state, find out possible values of n_1 and n_2.**

Sol. The time period T of an electron in a Bohr orbit of principal quantum number n is

$$T = \frac{n^3h^3}{4\pi^2K^2Z^2e^4m}$$

Thus $T \propto n^3$

$$\frac{T_1}{T_2} = \frac{n_1^3}{n_2^3}$$

As $T_1 = 8T_2$ the above relation gives

$$\left(\frac{n_1}{n_2}\right)^3 = 8$$

$\Rightarrow \quad n_1 = 2n_2$

Thus, possible values of n_1 and n_2 are:

$n_1 = 2$, $n_2 = 1$

$n_1 = 4$, $n_2 = 2$

$n_1 = 6$, $n_2 = 3$ and so on.

Q. **It is found experimentally that 13.6 eV energy is required to separate a hydrogen atom into a proton and an electron. Compute the orbital radius and the velocity of the electron in a hydrogen atom.**

Sol. From given data we can obseve that it is ionisation energy of hydrogen atom, now convert this energy into joule:

$$-13.6 \text{ eV} = -13.6 \times 1.6 \times 10^{-19} \text{ J} = -2.2 \times 10^{-18} \text{ J}$$

i.e. $E = -\dfrac{e^2}{8\pi \epsilon_0 r} = -2.2 \times 10^{-18} \text{J}$

This gives the orbital radius,

$$r = -\frac{e^2}{8\pi \epsilon_0 E} = -\frac{\left(9 \times 10^9 \text{Nm}^2/\text{C}^2\right)\left(1.6 \times 10^{-19}\text{C}\right)^2}{(2)\left(-2.2 \times 10^{-18}\text{J}\right)}$$

$$= 5.3 \times 10^{-11}\text{m}$$

The velocity of the revolving electron can be computed from figure with $m = 9.1 \times 10^{-31}$ kg

$$v = \frac{e}{\sqrt{4\pi \epsilon_0 mr}} = 2.2 \times 10^6 \text{ m/s}$$

Q. **It is given that the first excitation potential of a hypothetical hydrogen-like atom is 15 Volt. Calculate the third excitation potential of this hypothetical atom.**

Sol. Let energy of ground state = E_0

$E_0 = -13.6Z^2$ eV and $E_n = \dfrac{E_0}{n^2}$

$n = 2$, $E_2 = \dfrac{E_0}{4}$

Given $\dfrac{E_0}{4} - E_0 = 15$

$\dfrac{-3E_0}{4} = 15$

for $n = 4$, $E_4 = \dfrac{E_0}{16}$

Third excitation energy $= \dfrac{E_0}{16} - E_0 = -\dfrac{15}{16}E_0$

$$= -\frac{15}{16}\left(\frac{-4 \times 15}{3}\right)$$

$$= \frac{75}{4}\text{eV}$$

Third excitation potential is $\dfrac{75}{4}$V.

The Line Spectra of the Hydrogen Atom:

According to Bohr's third postulate when an electron moves from a higher energy state to a lower energy state a photon of energy equal to the energy difference of energy states is emitted.

$$h\nu = E_i - E_f$$

$$\nu = \frac{E_i - E_f}{h} = Z^2 Rc\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$$\nu = \frac{c}{\lambda} = Z^2 Rc\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$$\boxed{\bar{\nu} = \frac{1}{\lambda} = Z^2 R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)}$$

This is known as the Rydberg formula.

Where ; $R = \frac{me^4}{8\epsilon_0 h^3 c} = \left(1.097 \times 10^7\right) m^{-1}$ = Rydberg of constant

λ = wavelength

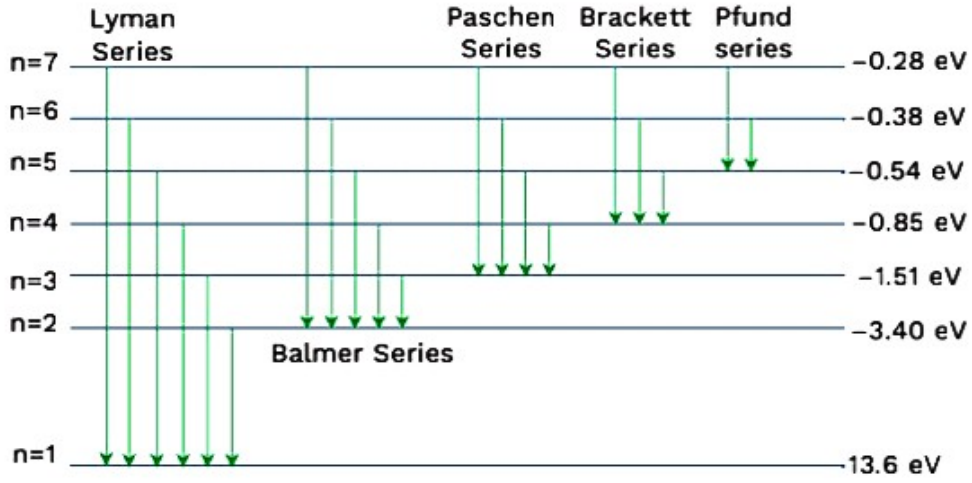
n_f = Final state

n_i = Initial state

The Hydrogen Atoms (Some Selected Lines):

Name of series	Number of Lines	Quantum Number			
		N_i (Lower state)	N_f (Upper state)	Wavelength (nm)	Energy
Lyman	I	1	2	121.6	10.2 eV
	II	1	3	102.6	12.09 eV
	III	1	4	97	12.78 eV
	Series Limit	1	∞ (Series Limit)	91.2	13.6 eV
Balmer	I	2	3	656.3	1.89 eV
	II	2	4	486.1	2.55 eV
	III	2	5	434.1	2.86 eV
	Series Limit	2	∞ (Series Limit)	364.6	3.41 eV
Paschen	I	3	4	1875	0.66 eV
	II	3	5	1281.8	0.97 eV
	III	3	6	1093.8	1.13 eV
	Series Limit	3	∞ (Series Limit)	822	1.51 eV

Series Limit: The minimum wavelength corresponding to maximum energy of a photon called series limit of that group.



For the Lyman Series $n_f = 1$, for Balmer Series $n_f = 2$ and so on.

Q. **Calculate the following parameters using the known value for hydrogen atom:**
(A) Find the Radius of the third orbit for Li^{+2} ion.
(B) Speed of electron in the fourth orbit for He^{+} ion.
(C) Calculate the angular momentum of an electron in the 3rd orbit of Helium ion

Sol. (A) Z = 3 for Li^{+2}, further we know that $r_n = \frac{n^2}{Z} a_0$

Put the value, n = 3, Z = 3 and $a_0 = 0.529$ Å
Radius for third orbit of Li^{+2}

$$= \frac{(3)^2}{(3)} \times 0.529\,\text{Å}$$

$$= 1.587\,\text{Å}$$

(B) Z = 2 for He^{+}

Also we know that $v_n = \frac{Z}{n} v_1$

Substituting n = 4, tomic no(Z) = 2 and velocity(v_1) = 2.19×10^6 m/s
We get, speed in fourth orbit,
For He^{+} $= \frac{Z}{n} v_1$

$$= \frac{2}{4} \times \left(2.19 \times 10^6\right) m / s$$

$$V_4 = 1.095 \times 10^6 \text{ m/s}$$

(C) $L_n = n\left(\frac{h}{2\pi}\right)$

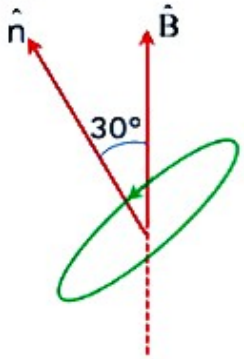
for n = 3

$$L_3 = 3 \times \left(\frac{h}{2\pi}\right)$$

Note: This result is independent of the value of (Z).

Q. **An electron is revolving in an anti-clockwise direction in the circular orbit of Radius R. The electron is in the ground state of the Hydrogen atom, as shown in the figure.**

(A) Find out an expression for the orbital magnetic dipole moment of the electron revolving in orbit.



(B) Suppose the atom is placed in a uniform magnetic field B such that the normal of the plane of the electron orbit makes an angle of 30° with the magnetic field. Calculate the Torque experienced by the orbiting electron..

Sol. According to Bohr's second postulate

$$mvr = \frac{nh}{2\pi} = \frac{h}{2\pi}$$

(As per, n = 1 first only)

$$v_1 = \frac{h}{2\pi m r_1}$$

According to the definition of current, the rate of flow of charge is called current; Hence current in the first orbit.

Now equation,

$$i = ev_1 = e\left(\frac{v_1}{2\pi r_1}\right)$$

$$= \frac{e}{2\pi r_1} \times v_1$$

$$i = \frac{e}{2\pi r_1} \times \frac{h}{2\pi m r_1} = \frac{eh}{4\pi^2 m r_1^2}$$

(A) Magnetic dipole moment,

$M_1 = i \times A_1$

$$= \frac{eh}{4\pi^2 m r_1^2} \times \pi r_1^2$$

$$\boxed{M_1 = \frac{eh}{4\pi m}}$$

(B) Torque on the orbiting electron in uniform magnetic field is:

$\vec{\tau} = \vec{M} \times \vec{B}$

$\tau = MB \sin 30^\circ$

$$= \frac{eh}{4\pi m} \times \frac{B}{2}$$

$$\boxed{\tau = \frac{ehB}{8\pi m}}$$

Q. **Determine the maximum wavelength that Hydrogen in its ground state can absorb. What would be the next smaller wavelength that would work?**

Sol. Maximum wave-length will correspond to the minimum energy transition of an electron from the ground state of the Hydrogen atom; the minimum energy transition is for n = 1 to n = 2, for which the absorbed energy photon required is

$\Delta E_{13} = E_3 - E_1$

$\Delta E_{13} = -1.5 \text{ eV} - (-13.6 \text{ eV})$

$\Delta E_{13} = 12.09 \text{ eV}$

If λ be its wavelength then it is given of

$$\frac{hc}{\lambda'} = 12.09 \text{eV}$$

$$\lambda' = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{12.09 \times 1.6 \times 10^{-19}}$$

$\lambda' = 1027.5 \text{ Å}$

Q. **Find kinetic energy, electrostatic potential energy, and total energy of a single electron in the 2nd excited state of Li^{+2} atom.**

Sol. $\because \quad E_1^H = -13.6\,eV$

further, $E \propto \frac{Z^2}{n^2}$

for $Li^{+2} \Rightarrow Z = 3$

and for 2nd excited state n = 3

$$E = -13.6\left(\frac{3}{3}\right)^2 = -13.6 \text{ eV}$$

$$K = |E| = 13.6 \text{ eV}$$

$$U = 2E = -27.2 \text{ eV}$$

In the above expression E is the total energy K is kinetic energy and U is the potential energy.

Q. **Calculate the potential energy, the kinetic energy, and total energy in the first and second orbit of the hydrogen atom if the potential energy in the first orbit is taken to be zero.**

Sol. $E_1 = -13.60$ eV $\qquad K_1 = -E_1 = 13.60$ eV

$$U_1 = 2E_1 = -27.20 \text{ eV}$$

$$E_2 = \frac{E_1}{(2)^2} = -3.40\,eV$$

$$K_2 = 3.40 \text{ eV}$$

$$U_2 = -6.80 \text{ eV}$$

Now $U_1 = 0$, i.e., potential energy has been increased by 27.20 eV. So, we will increase U and E in all energy states by 27.20 eV, while kinetic energy will remain unchanged. Changed values are-

orbit	K(eV)	U(eV)	E(eV)
first	13.60	0	13.60 eV
Second	3.40	20.40	23.80

Q. **Calculate the largest and shortest wavelengths in the Lyman series for the Hydrogen atom. Find out in which region of the electromagnetic spectrum does each series lies?**

Sol. The transition equation for Lyman series is given by:

$$\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{n^2}\right), \; n = 2, 3, \ldots\ldots$$

The largest wavelength is corresponding to n = 2

$$\frac{1}{\lambda_{max}} = 1.097 \times 10^7 \left(\frac{1}{1} - \frac{1}{4}\right) = 0.823 \times 10^7$$

$$\lambda_{max} = 1.2154 \times 10^{-7} m = 1215 \text{ Å}$$

The shortest wavelength corresponds to n = ∞

$$\frac{1}{\lambda_{min}} = 1.097 \times 10^7 \left(\frac{1}{1} - \frac{1}{\infty}\right)$$

$$\lambda_{min} = 0.911 \times 10^{-7} \text{ m} = 911 \text{ Å}$$

Both of these wave-lengths lie in the ultraviolet (U.V.) region of the electromagnetic spectrum.

Q. **The three energy levels L_1, L_2, and L_3 of the Hydrogen atom correspond to increasing energy values, i.e., $E_{L_1} < E_{L_2} < E_{L3}$. Suppose the wave-length corresponding to the transitions L_3 to L_2, L_2 to L_1, and L_3 to L_1 are λ_3, λ_2 and λ_1 respectively then:**

(A) $\lambda_3 = \lambda_1 + \lambda_2$ **(B)** $\frac{\lambda_3\lambda_2}{\lambda_2 + \lambda_3}$ **(C)** $\lambda_1 = \lambda_2 + \lambda_3$ **(D)** $\lambda_3 = \frac{\lambda_1\lambda_2}{\lambda_1 + \lambda_2}$

Sol.

$$\frac{1}{\lambda_1} = \frac{E_{L_3} - E_{L_1}}{\lambda_c} \quad \ldots(1)$$

$$\frac{1}{\lambda_2} = \frac{E_{L_2} - E_{L_1}}{\lambda_c} \quad \ldots(2)$$

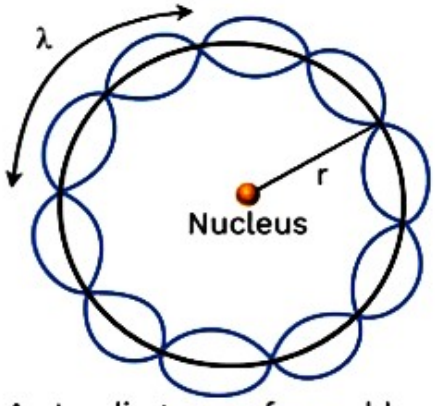
$$\frac{1}{\lambda_3} = \frac{E_{L_3} - E_{L_2}}{\lambda_c} \quad \ldots(3)$$

From equation (1), (2) and (3) we get,

$$\frac{1}{\lambda_2} + \frac{1}{\lambda_3} = \frac{1}{\lambda_1} \quad \Rightarrow \lambda_1 = \frac{\lambda_2\lambda_3}{\lambda_2 + \lambda_3}$$

De-Broglie's Explanation of Bohr's second postulate of quantization:

In 1924, De-Broglie proposed the concept of matter waves. Since the electron is a matter so he use the concept of matter-wave and explained the second postulate of Bohr's atomic model.



A standing wave formed by an orbital electron of an atom.

$$2\pi r_n = n\lambda \qquad \ldots(1)$$

Where; n = 1, 2, 3.....

i.e., the circumference of orbit should be integral multiple of de-Broglie wavelength of electron moving in n^{th} orbit.

Also, we know, de-Broglie wavelength = $\lambda = \frac{h}{mv}$ Therefore $2\pi r_n = \frac{h}{mv_n}$

$$\Rightarrow \quad \boxed{mv_n r_n = \frac{n \times h}{2\pi}} \qquad \ldots(2)$$

Angular momentum $n \times \left(\frac{h}{2\pi}\right)$.

This is 2nd postulates of Bohr's atomic model.

Effect of Motion of Nucleus on Bohr's Model (Double Star System) :-

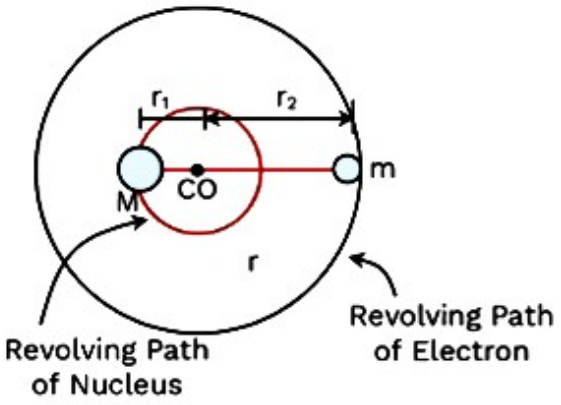
In a Hydrogen or Hydrogen like atom, we assume the nucleus at rest. We assume so because the mass of a nucleus is large compared to the mass of electron due to this center of mass of electron nucleus system lie very near to nucleus but practically nucleus and electron both revolve around the center of mass of electron nucleus system.

Effect of nucleus motion on energy of Atom:

Effect of Motion of Nucleus on Bohr's Model (Double Star System) :-

In a Hydrogen or Hydrogen like atom, we assume the nucleus at rest. We assume so because the mass of a nucleus is large compared to the mass of electron due to this center of mass of electron nucleus system lie very near to nucleus but practically nucleus and electron both revolve around the center of mass of electron nucleus system.

Derivation of Parameters :-



$r_1 = \left(\frac{m}{M+m}\right)(r)$; $r_2 = \left(\frac{M}{M+m}\right)(r)$ Charge on Nucleus = Ze

Charge on electron = e

Since both electron and nucleus revolve in a circular path. The centripetal force of the electrons is provided by electrostatic force.

$$\frac{k(Z)e^2}{r^2} = m_e \ \omega^2 \ r_2$$

$$\frac{kze^2}{r^2} = m\omega^2 \times \frac{Mr}{M+r}$$

$$\frac{kze^2}{r^2} = \frac{Mm}{(M+m)}(r\omega^2)$$

$$\frac{kze^2}{r^2} = (\mu)(r\omega^2) \qquad ...(1)$$

where $\mu = \frac{Mm}{M+m}$ = reduced mass.

Now use Bohr's 2nd postulate

Angular momentum of the system $= n\left(\frac{h}{2\pi}\right)$

$$I_1\omega_1 + I_2\omega_2 = n \times \frac{h}{2\pi}$$

$$\left(mr_2^2 + Mr_1^2\right) = n \times \frac{h}{2\pi}$$

$$\left[m \times \left(\frac{Mr}{M+m}\right)^2 + M\left(\frac{mr}{M+m}\right)^2\right] = n\frac{h}{2\pi}$$

after solving we will get:-

$$\left(\frac{mM}{M+m}\right)\omega r^2 = n \times \frac{h}{2\pi}$$

$$\boxed{\mu\left(\omega r^2\right) = n \times \frac{h}{2\pi}} \qquad \text{.....(2)}$$

Based on the above two equations we will get the following results,

(1) $r_n = 0.529\left(\frac{n^2}{z}\right)\left(\frac{m}{\mu}\right)$; m = mass of electron : μ = Reduced mass.

The further electrical potential energy of the system

$$U = \frac{-Ze^2}{4\pi\,\epsilon_c\, r} \quad \text{or} \quad U = \frac{-Z^2e^4\mu}{4\,\epsilon_0\, n^2h^2}$$

and kinetic energy, $K = \frac{1}{2}I\omega^2$

$$= \frac{1}{2}\mu r^2\omega^2 \Rightarrow K = \frac{1}{2}\mu v^2$$

$$K = \frac{Ze^2}{8\pi\,\epsilon_c\, r} = \frac{Z^2e^4\mu}{8\pi\,\epsilon_0^2\, n^2h^2}$$

∴ Total energy of system,

$E_n = K + U$

$$E_n = -\frac{\mu \cdot e^4}{8\,\epsilon_0^2\, n^2h^2}$$

This expression can also be written as:

$$E_n = -(13.6\,eV)\frac{Z^2}{n^2}\left(\frac{\mu}{m}\right)$$

(2) Total energy (T.E.) $= -13.6\frac{Z^2}{n^2}\left(\frac{\mu}{m}\right)$

(3) $v_n = 2.18\times 10^6\left(\frac{Z}{r}\right)$ m/sec ; Velocity expression is the same as previous.

Atomic Collision:

If the collision between two atoms or two particles is elastic, then the total kinetic energy of the system before a collision is the same as total energy after the collision. In this type of collision, there is no change in the state of atoms.

If a collision is inelastic, then there will be some loss of kinetic energy, which is used to excite one of the atoms.



Say e = 0



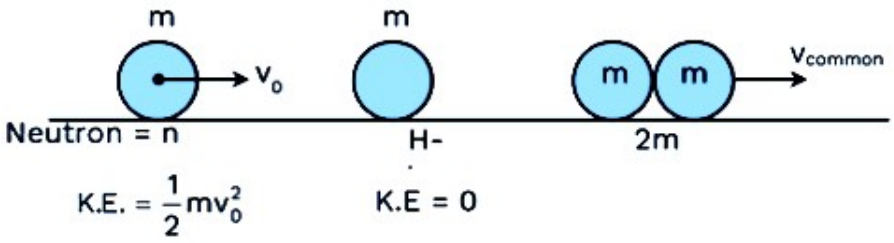
$$v = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}$$

Maximum possible loss in K.E.

$$= \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 - \frac{1}{2}(m_1 + m_2)v^2$$

$$= \frac{1}{2}\frac{m_1m_2}{m_1 + m_2}(u_1 - u_2)^2 < \text{maximum excitation energy.}$$

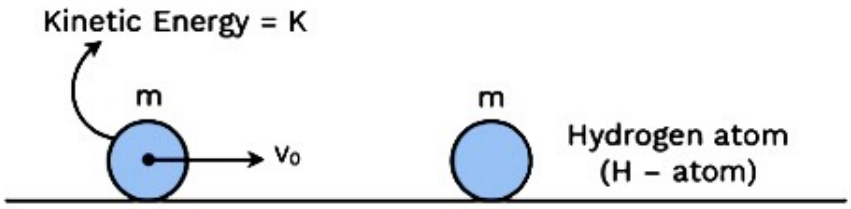
Q.



What will be the type of collision it is for given by different conditions.

(a) 0 < K < 20.4 eV **(b) K = 20.4 eV** **(c) K = 24.18 eV**

Sol.



$m \times v_0 + mv_0 = (2\ m)\ v_{common}$

$v_{comman} = \frac{v_0}{2}$ Final kinetic energy $= \frac{1}{2}(2m)(v_0/2)^2$

$= m \times \frac{v_0^2}{4}$

$(K.E)_f = \frac{mv_0^2}{4}$

Loss in kinetic energy = $(K.E)_f - (K.E.)_i$

$= \frac{mv_0^2}{4} - \frac{mv_0^2}{2}$

$= -\frac{mv_0^2}{4}$

$\boxed{\text{Loss} = -\frac{K}{2}}$

The loss in kinetic energy goes to excite the ground state electron. The minimum energy required to excite a ground-state electron = 10.2 eV

Case -1

If Loss < 10.2 eV $\Rightarrow$ No excitation

$$\left|\frac{K}{2}\right| < 10.2\,eV \quad \Rightarrow \quad |K| < 20.4\,eV$$

Conclusion if k/2 < 10.2, No excitation, loss can not be occur, that meance collision can not be inelastic. In this situation collision must be elastic

for, 0 < k < 20.4 eV ;

Collision must be elastic in nature.

Case – 2

For K = 20.4 ; In this case collision can be

(i) Perfectly elastic collision, it is not necessary that collision can occur.

(ii) Perfectly inelastic collision, in this case loss is maximum.

Case -3

Excitation energy required for 2nd state = 12.09 eV

So, k/2 = 12.09

$$\boxed{K = 24.18\,eV}$$

For, K = 24.18 eV, the following collision conditions are possible :-

(1) Perfectly elastic.

(2) Perfectly inelastic ($n_f = 3$, $n_i = 1$).

(3) Partially inelastic ($n_f = 2$, $n_i = 1$).

Answers :

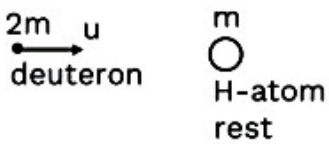
(a) Elastic

(b) Perfectly Elastic or Perfectly Inelastic.

(c) Perfectly elastic or perfectly inelastic or partially inelastic.

Q. **Calculate the minimum kinetic energy of deuteron so that it can excite H-atom.**

Sol.



$$K.E. = \frac{1}{2}2mu^2 = mu^2$$

Maximum possible loss in kinetic energy,

$$= \frac{1}{2}\frac{2m \cdot m}{2m+m}(u-0)^2$$

$mu^2 \geq 30.6$ eV

Minimum kinetic energy = 30.6 eV.

Q. **Suppose A neutron moving with speed v. It makes a head-on collision with a stationary hydrogen atom in the ground state. kept. Find the minimum kinetic energy of the neutron for which completely inelastic or partially inelastic collision may take place.**

Sol. Suppose the neutron and the hydrogen atom move at speeds v_1 and v_2 after the collision. The collision will be inelastic if a part of the kinetic energy is used to excite the atom.

Suppose an energy ΔE is used in this way. Using conservation of linear momentum and energy. $mv = mv_1 + mv_2$

$$\text{and } \frac{1}{2}mv^2 = \frac{1}{2}m_1^2 + \frac{1}{2}mv_2^2 + \Delta E \qquad ...(1)$$

$$\text{From (1), } v^2 = v_1^2 + v_2^2 + 2v_1v_2 \qquad ...(2)$$

$$\text{From (2), } v^2 = v_1^2 + v_2^2 + \frac{2\Delta E}{m}$$

$$\text{Thus, } 2v_1v_2 = \frac{2\Delta E}{m}$$

$$\text{Hence, } (v_1 - v_2)^2 - 4v_1v_2 = v^2 - \frac{4\Delta E}{m}$$

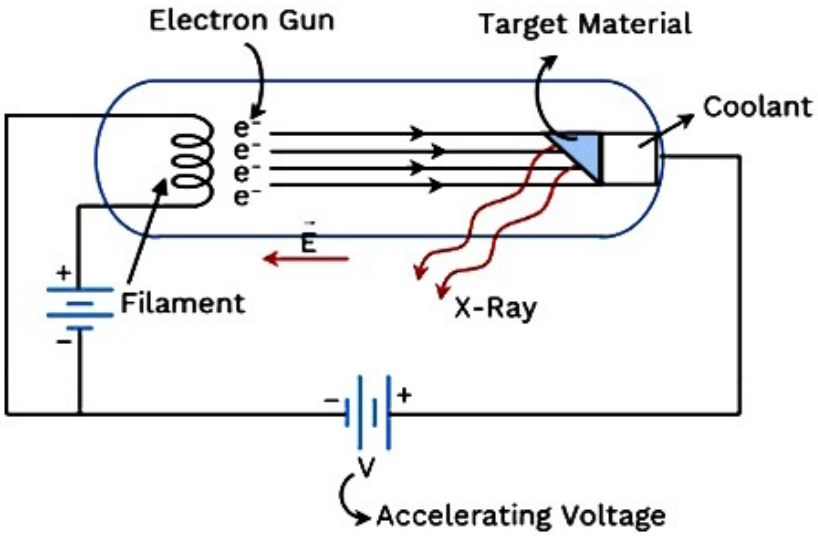
As of $v_1 - v_2$ must be real, $v^2 - \frac{4\Delta E}{m} \geq 0$ or $\frac{1}{2}mv^2 > 2\Delta E$.

The minimum energy that can be absorbed by the hydrogen atom in the ground state to go excited is 10.2 eV. Thus, the minimum kinetic energy of the neutron needed for an inelastic collision is

$$\frac{1}{2}mv_{min}^2 = 2 \times 10.2 \text{ eV} = 20.4 \text{ eV}.$$

X-Rays

X-Ray is a part of electromagnetic wave. It is discovered by WC roentgen in 1895. The wavelength of X-Rays found between 0.01 nm to 10 nm. . These rays are invisible to the eye, and their photons have energy around 1000 times more than visible light.
Production of X-Ray:- When a fast-moving electron strikes a target of high melting point and high atomic weight like tungsten, electromagnetic radiation is produced. This radiation is called X-Ray.



Coolidge tube for X-Ray Production

When voltage is applied across the filament, then it emits an electron. A collimator is used to give shape to the beam of electrons. Now when electrons strike the target material, then x-rays are produced, and during a collision, When electrons strike with the target, some part of the energy is lost & gets converted into heat since the target should not melt, so melting point and specific heat capacity should be high. Cu rod is attached to the target so that heat can be transferred. For more energetic electrons, the accelerating voltage is increased. For more number of photons, the voltage of filament is increased

Target material should have the following properties.

(1) High melting point.

(2) High specific heat capacity.

(3) High thermal conductivity.

(4) High atomic number.

Only about 1% of incident electron is energy is converted into x-Ray and the remaining 99% is converted into heat.

The intensity of X-Ray:- Greater is the number of electrons striking the target, move intensely are the X-ray.

Penetrating Power:- Energy of X-ray can be controlled by controlling the voltage across the filament and target material. High the potential difference

Q. **Suppose an X-ray tube operates at 20 kV. A particular electron loses 5% of its kinetic energy to emit on x-rays photon at the first collision. calculate wave-length corresponding to this photon.**

Sol. Kinetic energy acquired by the electron is

$$K = 20 \text{ keV} = 20 \times 10^3 \text{ eV}$$

The energy of the photon

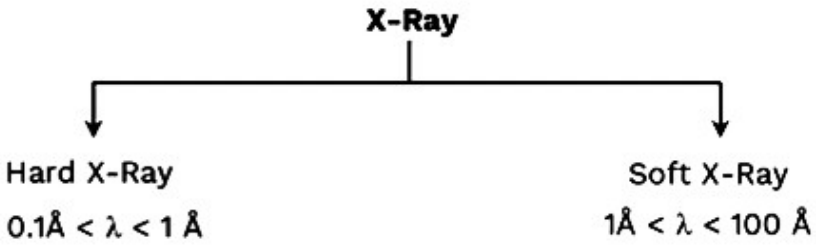
$$= 0.05 \times 20 \times 10^3 \text{ eV} = 10^3 \text{ eV}$$

Thus, $\frac{hc}{\lambda} = 10^3 eV$

$$\lambda = \frac{(4.14 \times 10^{-15} eV - s) \times (3 \times 10^8 m / s)}{10^3 eV}$$

$$= \frac{1242 \text{ eV} - nm}{10^3 eV} = 1.24 nm$$

Types of X-Rays:-



Q. **Find the energy, frequency, and momentum of an X-ray photon of wavelength 0.10 nm. Take hc = 12400 eV Å.**

Sol. Given wavelength of $\lambda = 0.1 \times 10^{-9}\ nm = 1Å$

Photon energy, $E = \frac{12400}{1} = 12.4\,KeV$

Frequency, $v = \frac{E}{h} = \frac{12.4 \times 10^3 \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}}$

$= 3 \times 10^{18}\ Hz$

Photon momentum, $p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{10^{-10}}$

$\Rightarrow \quad p = 6.63 \times 10^{-24}\ J.S.$

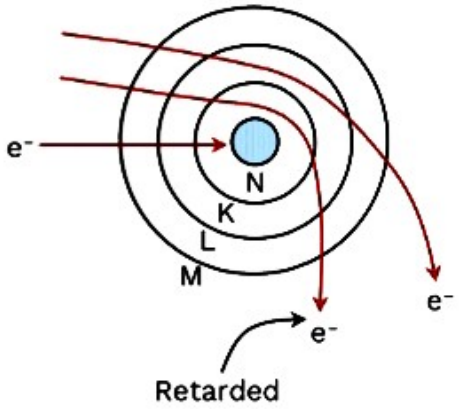
X-Ray Spectra Classification

(1). Continuous X-Rays

(2). Characteristic X-Rays

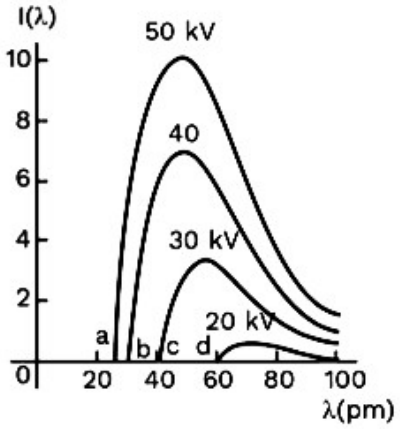
(1) Continuous X-Rays

These are produced due to the deacceleration of electrons passing near a heavy nucleus.



The loss in energy of electrons during retardation is emitted in form of a continuous X-ray.

Graph : I(λ) Vs wave length (λ)



Mathematical Relations:-

The maximum limiting frequency f_{max} or minimum limiting λ_{min} is obtained when the entire kinetic energy of the bombarding electron is converted to X-Ray.

Let : V_0 = Accelerating potential

$$\frac{1}{2}mv^2 = eV_0 = hf_{max}$$

$$\boxed{f_{max} = \frac{eV_0}{h}}$$

↓ Maximum frequency

From Relation, $eV_0 = \frac{hc}{\lambda_{min}}$

$$\lambda_{min} = \frac{hc}{eV_0}$$

$$\boxed{\lambda_{min} = \frac{12400}{V_0}}\ \text{Å}$$

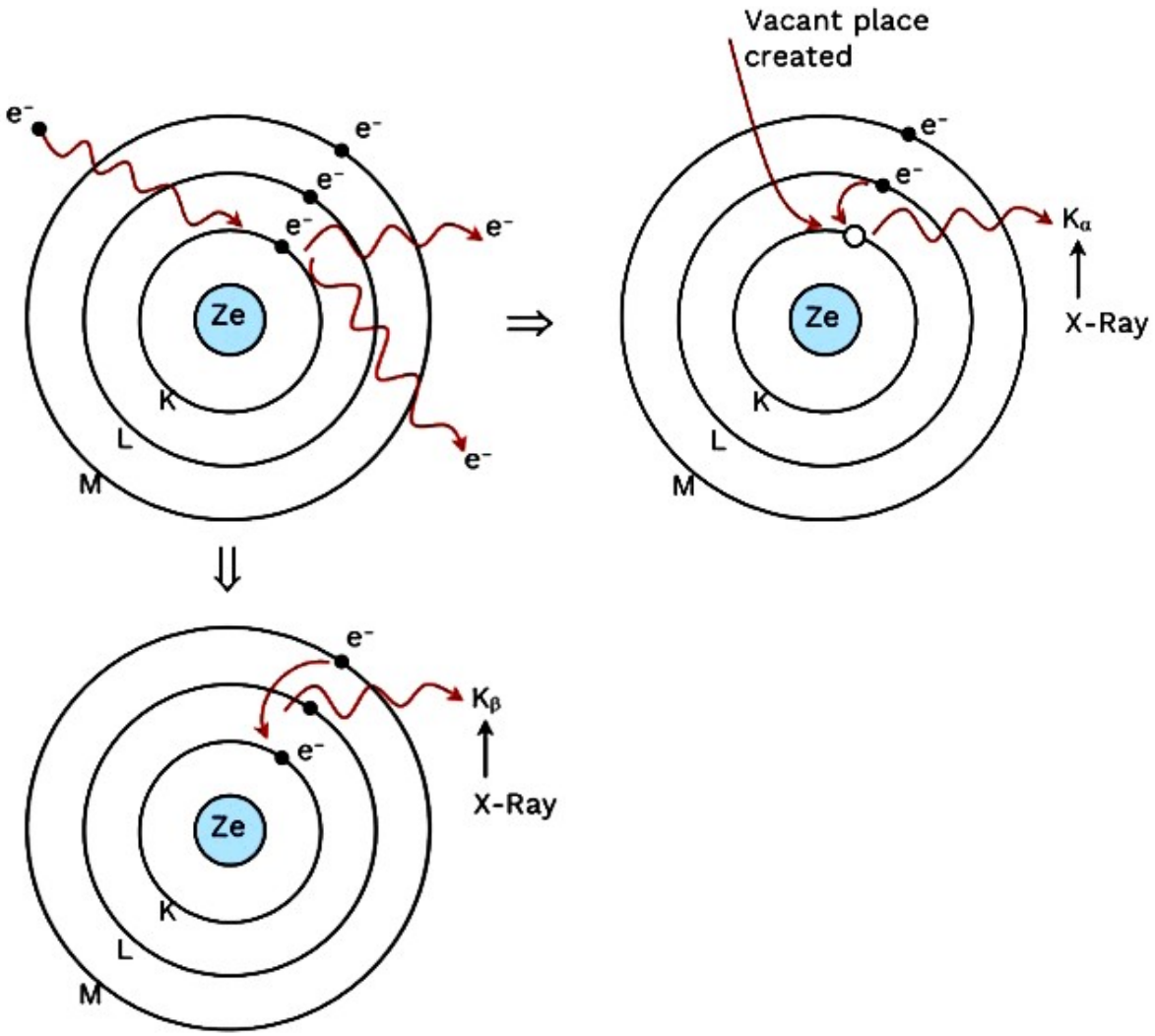
↳ Maximum Accelerating Voltage.

If K is the fraction of kinetic energy of electrons.

$$\boxed{\lambda = \frac{hc}{K(eV_0)}}$$

↳ Wavelength of continuous X-Ray.

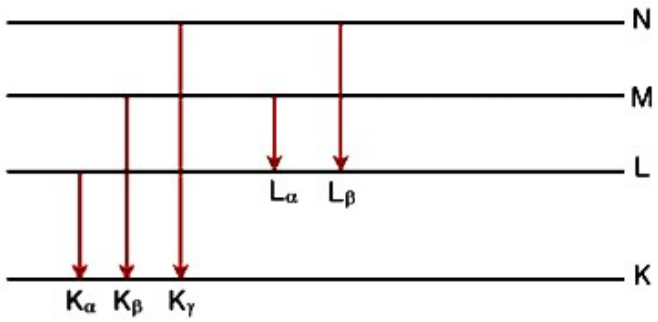
Characteristic X-Ray:



The line spectrum of X-Ray depends on the element used as the target. This line spectrum is called characteristic X-ray because it reflected the character of the target material. Every target material has a different characteristic X-Ray. Characteristics X-rays are produced when an electron knocks out an atomic electron from one of the inner levels. Now the create vacancy filled by L, M, N shells electrons.

When an electron jumps from higher energy orbit E_1 to lower energy orbit E_2, it radiates ($E_1 - E_2$) an amount of energy in form of an X-Ray.

If the transition is to the n = 1 level, the X-Rays are labeled K_α, K_β.... . If they are to the n = 2 level, they labeled L_α, L_β etc.



$$\lambda_{K_\alpha} = \frac{hc}{E_L - E_K}$$

$$\lambda_{L_\alpha} = \frac{hc}{E_M - E_L}$$

Graph - Intensity versus wavelength (λ) curve:-

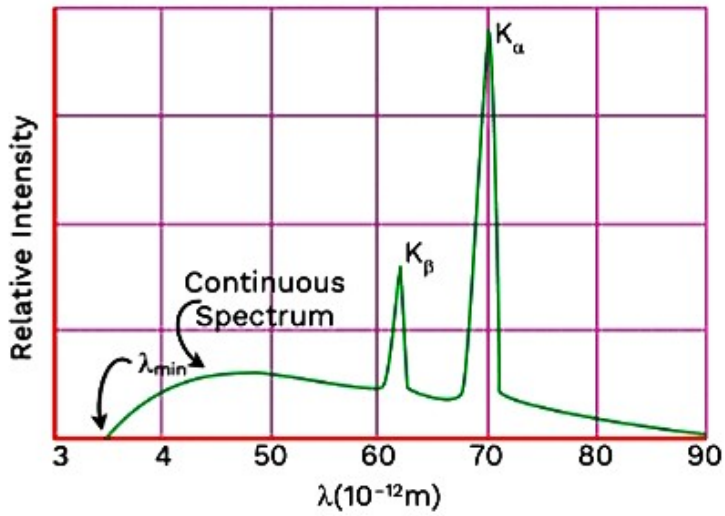


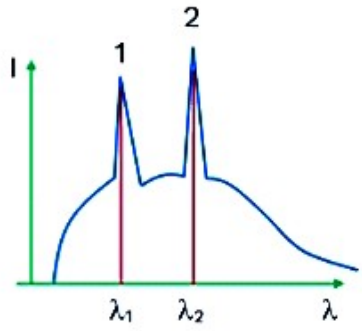
Fig. Characteristic X-Ray curve

The sharp peaks obtained in the graph are characteristic X-rays because they are characteristic of the target material.

λ_1, λ_2, λ_3, λ_4 ..... = characteristics

The wave-length of material having atomic number Z is called characteristic X-ray, and the spectrum obtained is called characteristics spectrum. If the target of atomic number Z is used then, peaks are shifted.

Q. **Find which is K_α and K_β.**



Sol. $$\Delta E = \frac{hc}{\lambda}$$

$$\lambda = \frac{hc}{\Delta E}$$

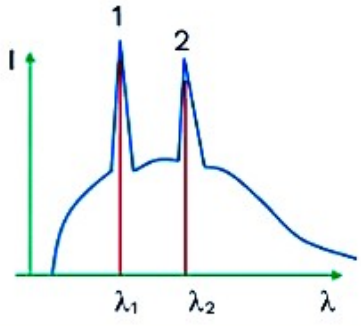
Since the energy difference of K_α is less than K_β,

$$\Delta E_{K_\alpha} < \Delta E_{K_\beta}$$

$$\lambda_{K_\beta} < \lambda_{K_\alpha}$$

Peak no 1 for K_β and peak no 2 for k_α.

Q.



Find which is L_α and L_α.

Sol. $\Delta E_{K_\alpha} > \Delta E_{L_\alpha}$

Peak 1 for k_α and peak 2 for L_α.

Moseley's Law:

In 1913, Moseley measured the frequency of characteristic X-rays for a large number of elements and plotted the square root of frequencies almost against 'Z.' He discovered that plot is the almost straight line not passing through the origin.

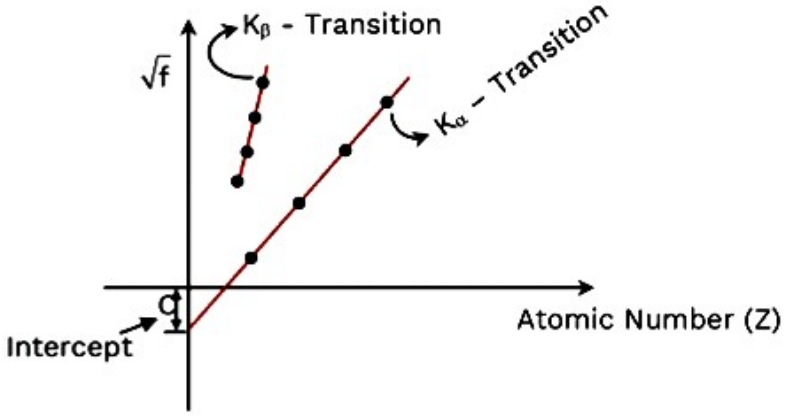
According to Mosely Law:

"The square root of the frequency of the emitted X-Ray is approximately proportional to the 'atomic number".

$$\sqrt{f} \propto (Z)$$

Graph:-

Mosely plotted the square root of the frequency with atomic number (Z) for K_α – transition.



From above graph Mosely find a relation –

$$y = mx + C$$

$$\sqrt{f} = a(Z) + (-C)$$

$$\sqrt{f} = a\left(Z - \frac{C}{a}\right)$$

$$\boxed{\sqrt{f} = a(Z - b)}$$

where a and b are constant t.

b = 1 for Kα series. It depends on the nature of the target material.

For hydrogen-like atoms we use relation

$$\boxed{\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)}$$

For K_α - Line :-

$n_1 = 1 : n_2 = 2$

$$\frac{1}{\lambda} = RZ^2\left(\frac{1}{1^2} - \frac{1}{2^2}\right)$$

$$= RZ^2\left(1 - \frac{1}{4}\right)$$

$$= RZ^2\left(\frac{4-1}{4}\right)$$

$$\frac{1}{\lambda} = \frac{3RZ^2}{4}$$

$C = f \times \lambda$

$$\frac{f}{C} = \frac{3RZ^2}{4}$$

$$f = \frac{3RC}{4}Z^2$$

$$\sqrt{f} = \left(\sqrt{\frac{3RC}{4}}\right)(Z)$$

If we consider screening constant for K_α transition-

$$\boxed{\sqrt{f} = \sqrt{\frac{3RC}{4}}(Z-1)}$$

$$\sqrt{f} = \sqrt{\frac{3RC}{4}}(Z-1)$$

Where a and b are constant for a particular shell and material (Z)

$$a = \sqrt{\frac{3RC}{4}}$$

b = 1 ; for K_α transition

R = Rydberg constant

Note : b = 1 for K_α - Series

b = 7.4 for L_α - Series

b = 19.2 for M_α – Series.

Wavelength distribution of X-Ray production in a molybdenum target.

Q. **Use Moseley's law with b = 1 to find the frequency of the K_α x-rays of La (Z = 57), if the frequency of the K_α x-rays of Cu (Z = 29) is known to be 1.88×10^{18} Hz.**

Sol. Using the equation:

$$\sqrt{f} = a(Z - b) \quad (b = 1)$$

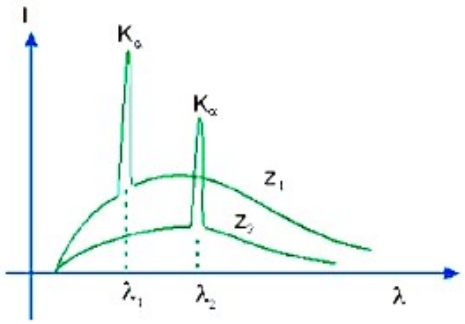
$$\frac{f_{La}}{f_{cu}} = \frac{(Z_{La} - 1)^2}{(Z_{cu} - 1)^2}$$

$$f_{La} = f_{cu}\left(\frac{Z_{La} - 1}{Z_{cu} - 1}\right)^2$$

$$= 1.88 \times 10^{18}\left(\frac{57 - 1}{29 - 1}\right)^2$$

$$f_{La} = 7.52 \times 10^{18} \text{ Hz}$$

Q.



Find in Z_1 and Z_2 which one is greater.

Sol. $\because \quad \sqrt{\nu} = \sqrt{cR\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)}.(Z - b)$

If Z is greater, then ν will be greater, λ will be less

$\therefore \quad \lambda_1 < \lambda_2$

$\therefore \quad Z_1 > Z_2$.

Examples Based on the Atomic Structure

Q. **How many times does the electron go round the first Bohr orbit of a hydrogen atom in 1 s?**

Sol. We know the frequency of revolution of electron is:

$$f_n = \frac{V_n}{2\pi r_n}$$

It is the number of revolutions in 1 second it can be given as for n = 1 orbit of hydrogen atom as.

$$f_1 = \frac{2.18 \times 10^6}{2 \times 3.14 \times 0.529 \times 10^{-10}} (\text{sec}^{-1})$$

$$f_1 = 6.56 \times 10^{15} (\text{sec}^{-1})$$

Q. **A Hydrogen atom in a state having Binding energy of 0.85 eV makes the transition to a state with excitation energy of 10.2 eV.**
(A) Find the Energy and wavelength of a photon emitted.
(B) Show the transition on an energy level diagram for Hydrogen indicating the quantum number.

Sol. (A) Binding energy $= \frac{13.6}{n_1^2} = 0.85\,\text{eV} = \frac{13.6}{n_1^2}$ where n = 4

$$E_{n_1} = -0.85\,\text{eV}$$

Suppose the transition takes place to state n_2, for which excitation energy is 10.2 eV. In these cases, ground state energy + excitation energy = E_{n_2}

$$-13.6 + 10.2 = \frac{13.6}{n_2^2}$$

Solving we get $n_2 = 2$ and we have,

$$E_{n_2} = -3.4\text{eV}$$

Energy of photon emitted = $E_{n_1} - E_{n_2}$

$$= 3.4 - 0.85 = 2.55 \text{ eV}$$

Now the wavelength emitted is:

$$\lambda = \frac{hc}{E}$$

$$\lambda = \frac{12431}{2.55}\text{Å}$$

$$\lambda = 4875\ \text{Å}$$

(B) Energy level diagram is shown in the figure.



Q. **Electrons of energies 10.20 eV and 12.09 eV can cause radiation to be emitted from hydrogen atoms. Calculate in each case the principal quantum number of the orbit to which electron in the hydrogen atom is raised and the wavelength of the radiation emitted if it drops back to the ground state. Take hc = 12375 eVÅ**

Sol. Since the orbital energy of an electron revolving in n^{th} orbit is given by

$$E_n = -\left(\frac{13.6}{n^2}\right)\text{eV}$$

When n = 1, E_1 = –13.6 eV

n = 2, E_2 = –3.4 eV

n = 3, E_3 = –1.51 eV

Here we observe that 10.20 eV = $E_2 - E_1$ and 12.09 eV = $E_3 - E_1$.

So, by absorbing a radiation photon of 10.2 eV the electron will make a transition to n = 2 state and by absorbing a 12.09 eV photon the electron will make a transition to n = 3 state. Now after the lifetime of excited states, the electron in n = 2 and n = 3 will make transitions to lower states and ultimately come back to ground state. In this process, the possibility of transition are

n = 3 to n = 2

n = 3 to n = 1

n = 2 to n = 1

In the above three transitions, the amount of energy released will be

$$\Delta E_{32} = (-1.51\text{eV}) - (-3.4\,\text{eV}) = 1.89\,\text{eV}$$

$$\Delta E_{31} = (-1.51\text{eV}) - (-13.6\,\text{eV}) = 12.09\,\text{eV}$$

$$\Delta E_{21} = (-3.4\,\text{eV}) - (-13.6\,\text{eV}) = 10.20\,\text{eV}$$

Thus, the wave-length of radiations of the corresponding transition is

$$\lambda_{32} = \frac{12375}{1.89} = 6548\text{Å}$$

$$\lambda_{31} = \frac{12375}{12.09} = 1024\text{Å}$$

$$\lambda_{21} = \frac{12375}{10.2} = 1213\text{Å}$$

Q. **Suppose the wave-length of the nth line of the Lyman series is equal to the de-Broglie wave-length of an electron in the initial orbit of a hydrogen-like element (Z = 11). Find the value of n.**

Sol. n^{th} line of the Lyman series means the transition from $(n + 1)^{th}$ state to the first state.

$$\frac{1}{\lambda} = RZ^2\left(1 - \frac{1}{(n+1)^2}\right) \qquad ...(1)$$

de-Broglie wavelength in $(n + 1)^{th}$ orbit is

$$\lambda = \frac{h}{mv} = \frac{hr}{mvr} = \frac{(2\pi)(hr)}{(n+1)h} = \frac{2\pi r}{(n+1)}$$

$$\Rightarrow \quad \frac{1}{\lambda} = \frac{(n+1)}{2\pi r} \qquad ...(2)$$

Equating equations (i) and (ii), we get

$$\left(\frac{n+1}{2\pi r}\right) = RZ^2\left(\frac{n(n+2)}{(n+1)^2}\right) \qquad ...(3)$$

Since, $r \propto \frac{n^2}{Z}$

$$\Rightarrow \quad r = \frac{(n+1)^2}{11} r_0$$

Substituting in equation (iii), we get

$$\frac{11}{2\pi r_0} = \frac{R(11)^2(n)(n+2)}{(n+1)}$$

$$\Rightarrow \quad (n+1) = (1.09 \times 10^7)(11)(2\pi) \times (0.529 \times 10^{-10})(n^2 + 2n)$$

Solving this equation, we get N = 24.

Q. **The emission spectrum of hydrogen atoms has two lines of Balmer series with wave-length 4102 Å and 4861 Å. To what series does a spectral line belong if its wave number is equal to the difference of wavenumbers of the above two lines? What is the wavelength of this line? [Take R = 1.097 × 10^7 m^{-1}].**

Sol. According to given problem:

$$\frac{1}{\lambda_1} = RZ^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$$\Rightarrow \quad \frac{1}{\lambda_1} = RZ^2\left[\frac{1}{2^2} - \frac{1}{n_2^2}\right] \qquad ...(1)$$

$$\text{and} \quad \frac{1}{\lambda_2} = RZ^2\left[\frac{1}{2^2} - \frac{1}{(n_2')^2}\right] \qquad ...(2)$$

$$\Rightarrow \quad \frac{1}{\lambda_1} - \frac{1}{\lambda_2} = \left[\frac{1}{(n_2')^2} - \frac{1}{n_2^2}\right]RZ^2$$

The wavenumbers of above radiations are

$$\bar{\nu}_1 = \frac{1}{\lambda_1} \text{ and } \bar{\nu}_2 = \frac{1}{\lambda_2}$$

$$\text{Thus, } \bar{\nu}_1 - \bar{\nu}_2 = \left[\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right]$$

$$\Rightarrow \quad \bar{\nu}_1 - \bar{\nu}_2 = RZ^2\left[\frac{1}{(n_2')^2} - \frac{1}{n_2^2}\right]$$

Let the wave number of the third emitted line will be

$$\bar{\nu} = \bar{\nu}_1 - \bar{\nu}_2$$

$$\Rightarrow \quad \frac{1}{\lambda} = RZ^2\left[\frac{1}{(n_2')^2} - \frac{1}{n_2^2}\right] \qquad ...(3)$$

From equation (1), we have

$$\frac{1}{RZ^2\lambda_1} = \frac{1}{4} - \frac{1}{n_2^2}$$

$$\Rightarrow \quad \frac{1}{n_2^2} = \frac{1}{4} - \frac{1}{RZ^2\lambda_1}$$

$$\Rightarrow \quad \frac{1}{n_2^2} = \frac{1}{4} - \frac{1}{(1.097 \times 10^7)(1)^2(4102 \times 10^{-10})}$$

Solving for n_2, we get $n_2 = 6$ similarly, from equation (2), we have $n_2' = 4$.

From so, the third line belongs to the Brackett series because the transition is from level to 4. The wavelength is given by equation (3) so,

$$\frac{1}{\lambda} = (1.097 \times 10^7)(1)^2\left[\frac{1}{(4)^2} - \frac{1}{(6)^2}\right]$$

$$\Rightarrow \quad \lambda = 2.62 \times 10^{-6}\text{m}.$$

Q. **Find the constants a and b in Moseley's equation $\sqrt{\nu} = a(Z - b)$ from the following data.**

Element	Z	Wavelength of Kα X-ray
Mo	42	71 pm
Co	27	178.5 pm

Sol. Moseley's equation is $\sqrt{\nu} = a(Z - b)$

Thus, $\sqrt{\frac{c}{\lambda_1}} = a(Z_1 - b)$ ...(1)

and $\sqrt{\frac{c}{\lambda_2}} = a(Z_2 - b)$ ...(2)

From (1) and (2) $\sqrt{c}\left(\frac{1}{\sqrt{\lambda_1}} - \frac{1}{\sqrt{\lambda_2}}\right) = a(Z_1 - Z_2)$ or $a = \frac{\sqrt{c}}{(Z_1 - Z_2)}\left(\frac{1}{\sqrt{\lambda_1}} - \frac{1}{\sqrt{\lambda_2}}\right)$

$$= \frac{(3\times10^{8}\,m/s)^{1/2}}{42-27}\left[\frac{1}{(71\times10^{-12}m)^{1/2}} - \frac{1}{(178.5\times10^{-12}m)^{1/2}}\right] = 5.0\times10^{7}(Hz)^{1/2}$$

Dividing (1) by (2)

$$\sqrt{\frac{\lambda_2}{\lambda_1}} = \frac{Z_1 - b}{Z_2 - b} \text{ or } \sqrt{\frac{178.5}{71}} = \frac{42-b}{27-b} \text{ or } b = 1.37$$